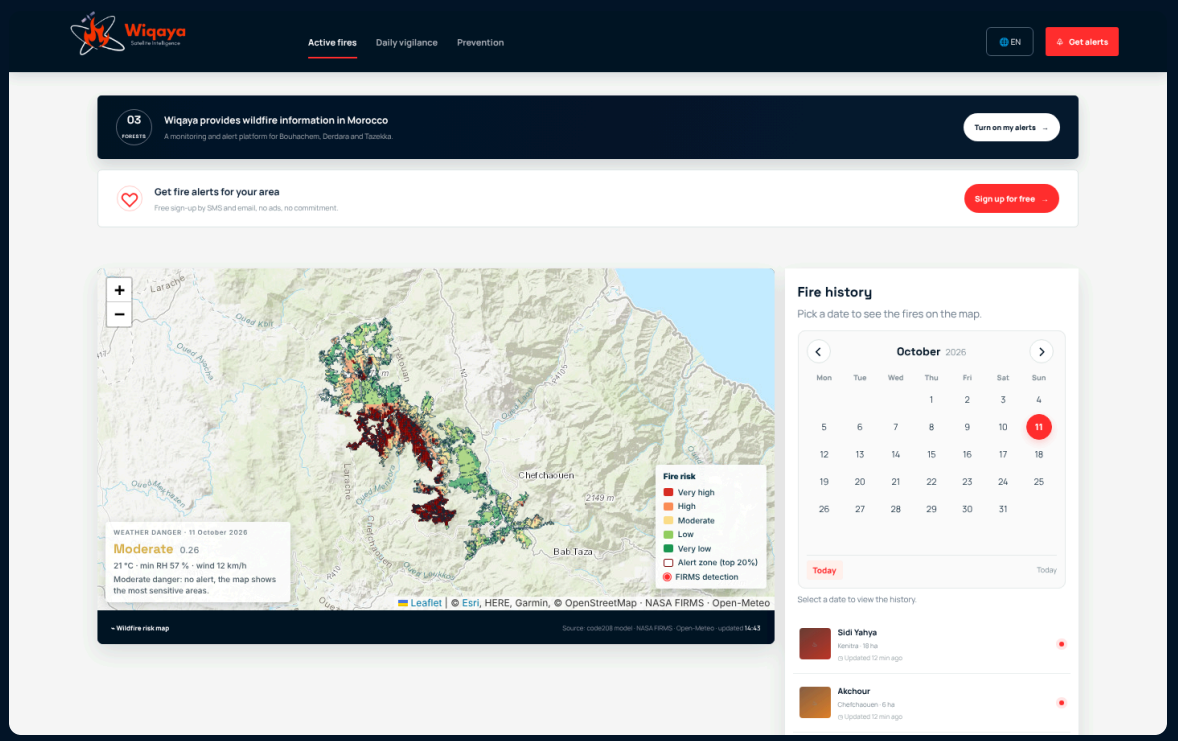
# Wiqaya

## Wildfire risk mapping and alerts for the forests of northern Morocco

Satellite fire history + terrain + population → a 250 m fire-susceptibility map, a daily weather danger index and trilingual citizen alerts.

| | |
|---|---|
| Team | **Wiqaya**: Sanae Belfrouh, Wiame Kane |
| Country | **Morocco** |
| Theme | **Climate & Disaster Resilience** |
| Study area | Bouhachem, Derdara and Jbel Sougna forests (25,900 ha) |



THE PROBLEM

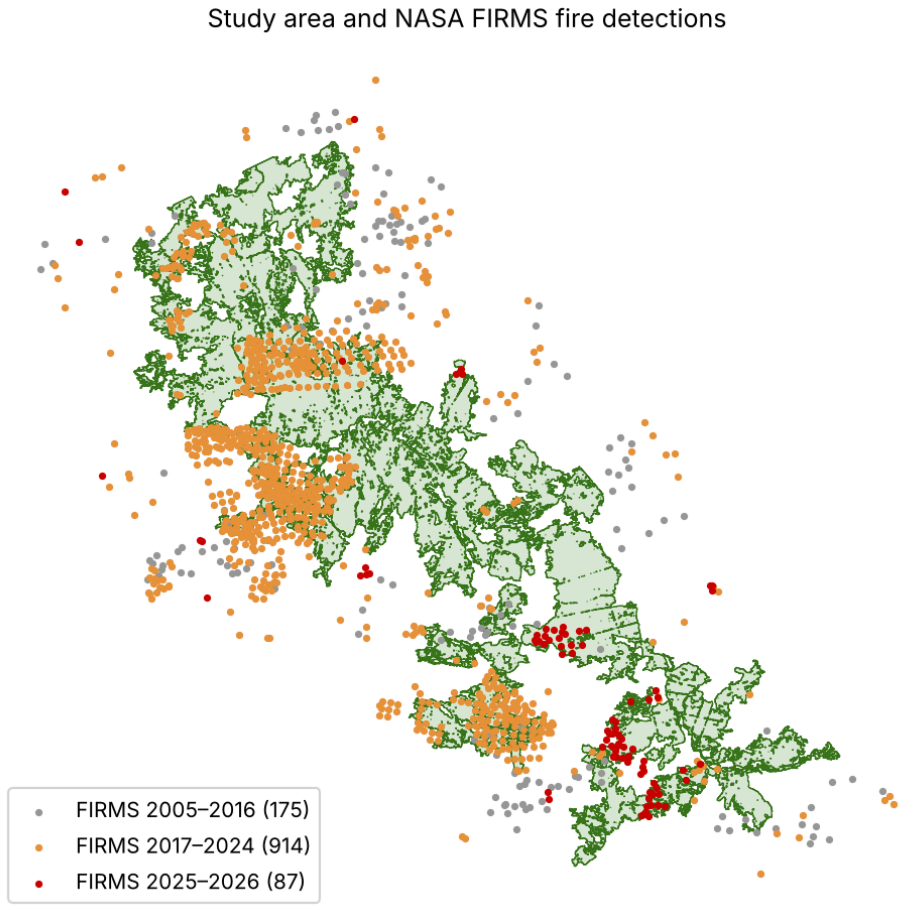
# The western Rif burns almost every summer, and the warning comes too late

- **Recent large fires:** Bouhachem 2022 (> 7,500 ha), Larache 2022 (> 10,500 ha), Derdara 2021 and Bouhachem 2020 (~1,100 ha each), two new fires in August 2025.
- **Drivers:** hot, dry *chergui* winds, steep slopes, and villages (*douars*) inside the forest.
- **The gap:** danger bulletins are province-scale, and satellite detections arrive **after** ignition. Nobody can say *which 250 m cells* need attention *today*.
- **Why satellites:** NASA FIRMS gives the only consistent 20-year fire history of these remote mountains (MODIS since 2005, VIIRS 375 m since 2017).

**10,600**
people living inside the forest (WorldPop 2017)

**264,000**
people within 10 km

**445**
FIRMS detections inside the forest since 2005




Study area and NASA FIRMS fire detections by period (notebook output).

BUSINESS USE CASE

# Who uses Wiqaya, and for which decision

| | |
|---|---|
| **End users** | Forest managers (national forestry agency ANEF), Civil Protection, provinces and communes. Residents of the villages inside and around the forest. |
| **Their decision** | Every summer morning: **where** to send patrols, watch-tower crews and first-intervention teams, and **whether** today is dangerous enough to warn residents and restrict burning and works. |
| **What they use today** | Province-scale danger bulletins, field knowledge, and fire detections (FIRMS, EFFIS) that come **after** ignition. No cell-level daily view; no alerts in Arabic and French. |
| **What Wiqaya adds** | A 250 m risk map with alert zones and an explanation per cell, a daily weather danger index, a 20-year fire history calendar, SMS / email sign-up in Arabic, French and English, and an open GeoJSON API. |
| **Business model** | Subscription per protected area for institutions (dashboard, API, alert dispatch). Free citizen alerts, funded by the institutional contracts or by CSR partners. Built on open data, so there is no imagery cost. |

DATA

# Open Earth-observation data, all documented and cited

| Dataset | Product · provider | Dates | Use |
|---|---|---|---|
| **Active fires** | NASA FIRMS: MODIS C6.1 and VIIRS S-NPP 375 m | 2005 → 2026 · 1,176 detections | Labels, fire history, history map |
| **Terrain** | ≈30 m digital elevation model | static | Elevation, slope, aspect |
| **Population** | WorldPop `mar_ppp_2017` (CC BY 4.0) | 2017 | Exposure per cell |
| **Weather** | Open-Meteo: ERA5 archive and forecast (CC BY 4.0) | 2017 → today | Daily weather danger index |
| **Optical** (research notebook) | Sentinel-2 L2A and Landsat C2 L2 via Planetary Computer | 2005 → 2025 | NDVI, NDMI, NBR, dNBR burned areas |
| **Land cover, OSM** (research notebook) | 2021 classification · OpenStreetMap 2022 | 2021–2022 | Fuel, human pressure |

**Hyperspectral data:** not used in this PoC. It is the first item of the incubation roadmap: EnMAP or Arab Satellite 813 to map **live fuel moisture** and **vegetation species** at cell level, the two drivers our current data see only indirectly.

APPROACH

# Workflow, from satellite detections to alert zones

**1 · Inputs**
FIRMS points 2005–2026
DEM ≈30 m
WorldPop 2017
Forest boundaries

→

**2 · Grid and features**
6,071 cells of 250 m
elevation, slope, aspect (circular mean), population

→

**3 · Fire history**
FIRMS within 375 m
years since last fire
strictly before the date (no leakage)

→

**4 · Model selection**
yearly panel 2017–2024
logistic regression vs random forest
leave-one-year-out

**5 · Fuel rule**
0 if burned < 1 year ago
× 0.5 if burned 1–5 years ago

→

**6 · Blind test**
retrain 2017–2024
score the 2025 fires (never seen)

→

**7 · Risk map**
susceptibility per cell
5 KMeans classes
alert = riskiest 20 %

+

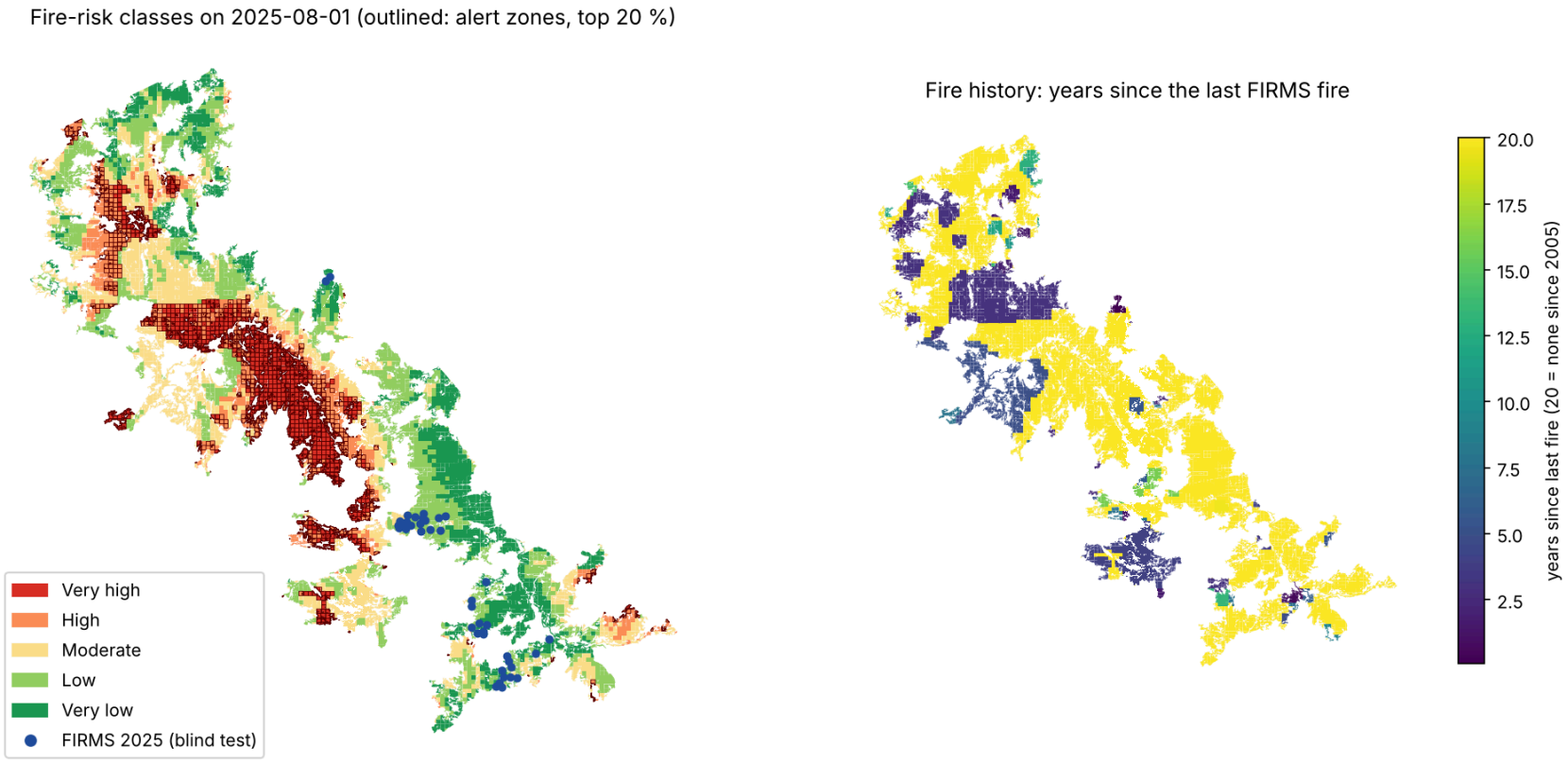
**8 · Weather danger**
max T, max VPD, max wind, min RH
ranked vs 2017–2024 summers
low → very high

Reproducible: `notebooks/02_main_analysis.ipynb` runs end to end in about 5 minutes from the committed sample inputs, with fixed seeds and relative paths. The research notebook adds a dNBR-trained "fire day" model with 29 inputs.

EXAMPLE OUTPUT · NOTEBOOK

# Fire-risk classes on 1 August 2025, before the 2025 fires

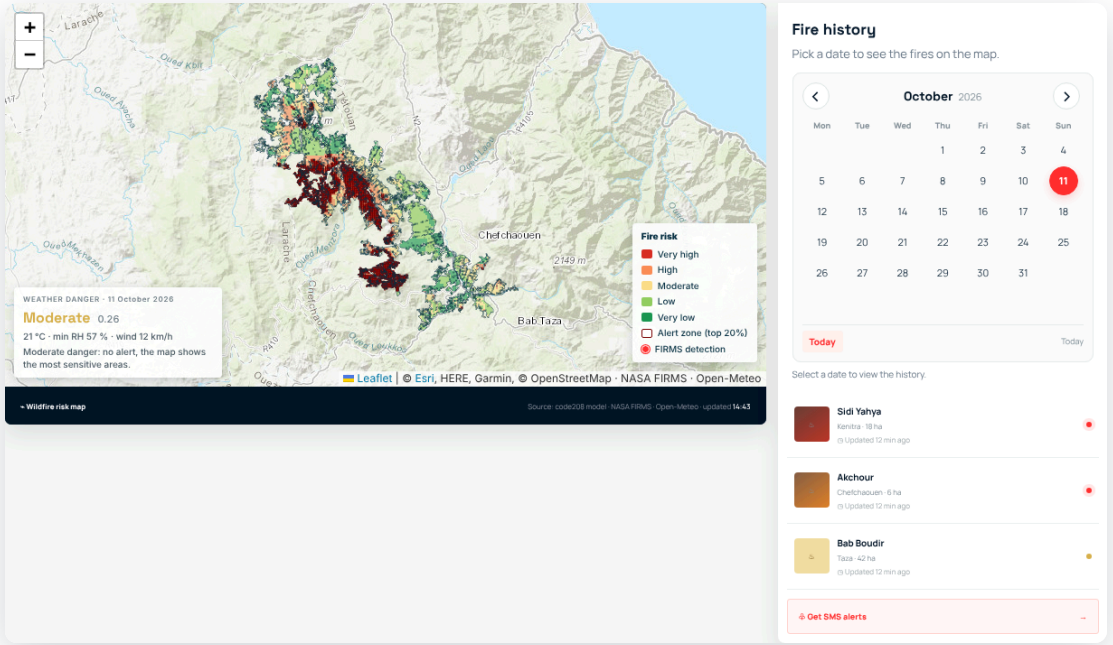
Wiqaya — wildfire susceptibility, Bouhachem / Derdara / Jbel Sougna · model: Logistic regression · blind-test ROC-AUC 2025 = 0.25



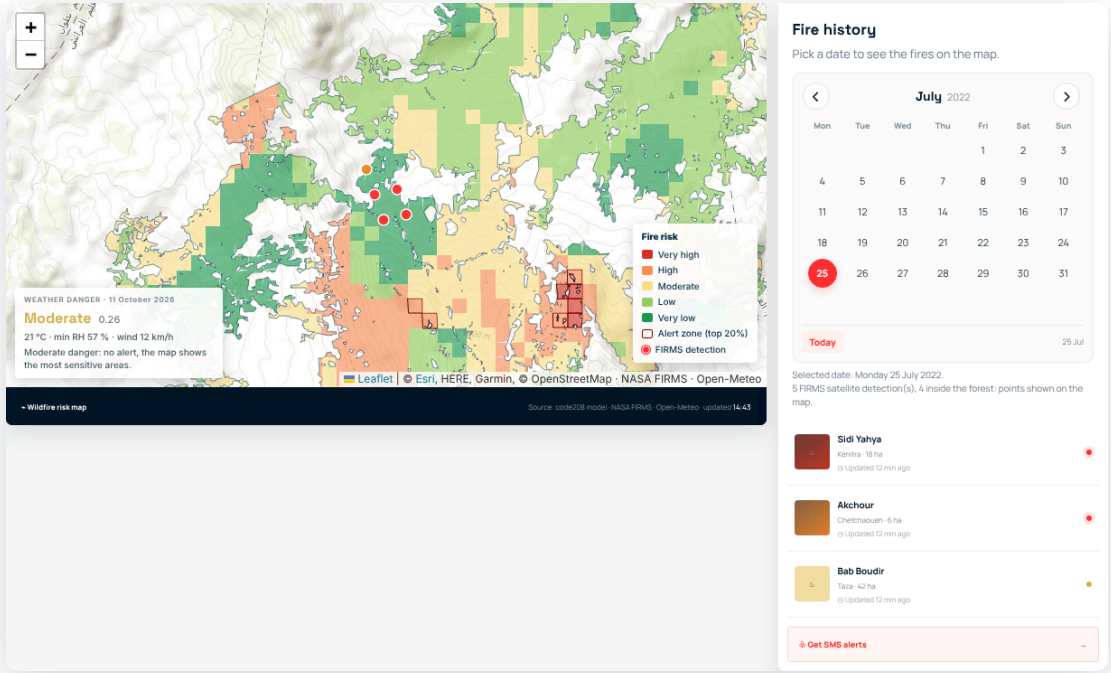

Left: 5 risk classes, with the alert zones (riskiest 20 % of the forest) outlined. Blue points: the 2025 FIRMS fires, used as a blind test. Right: years since the last FIRMS fire (20 = none since 2005). Output `results/example_output.png`.

EXAMPLE OUTPUT · WEB PLATFORM

# The same model, delivered to users



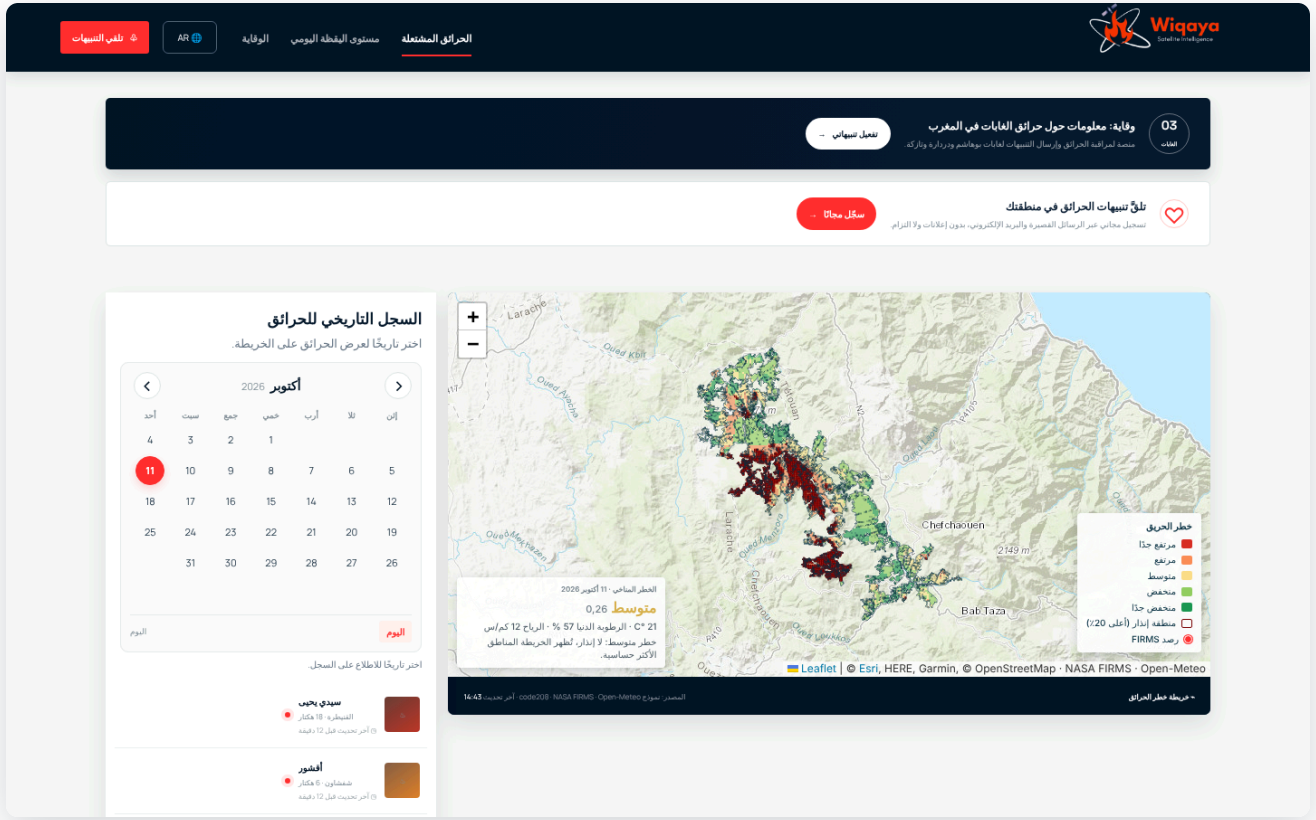

Daily risk map with the weather danger of the day and the 5-class legend.



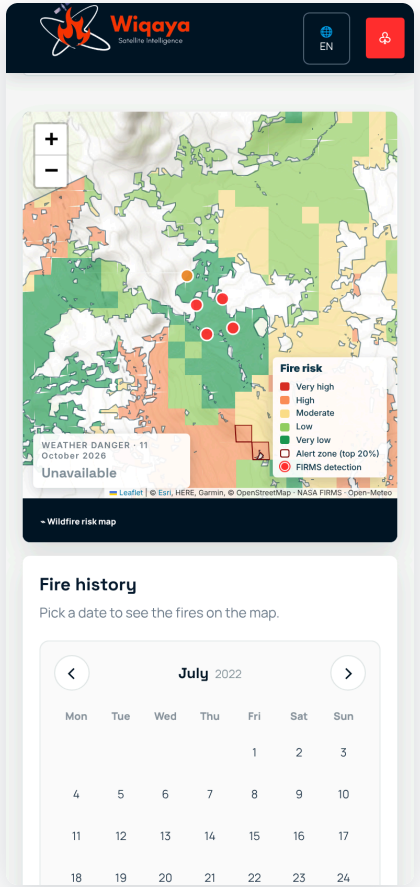

Fire history: on 25 July 2022, the start of the Bouhachem fire, FIRMS shows 5 detections, 4 inside the forest.

EXAMPLE OUTPUT · FOR RESIDENTS

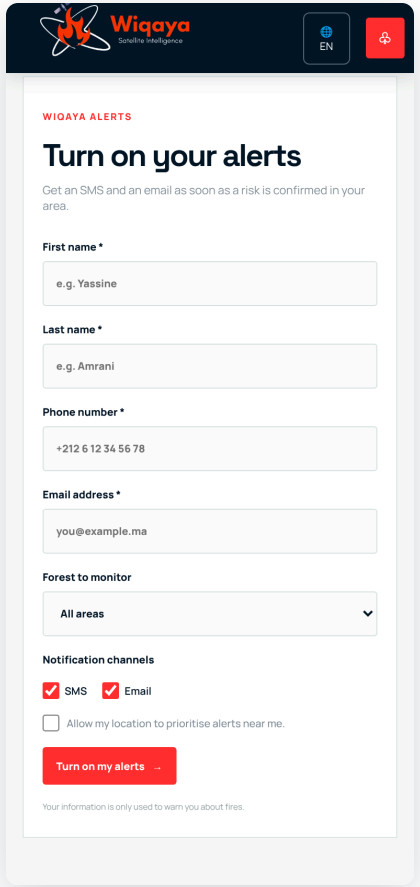
# Arabic, French and English, on any phone




Full right-to-left Arabic interface.




Fire history on a phone.




SMS / email alert sign-up.

VALIDATION AND LIMITATIONS

# A working, reproducible pipeline, not yet a reliable predictor

| Test | Result |
|---|---|
| Leave-one-year-out 2017–2024, logistic regression (selected) | mean ROC-AUC per year **0.62** · global 0.65 · PR lift 1.64 |
| Leave-one-year-out, random forest | mean ROC-AUC per year 0.56 · global 0.60 |
| Per held-out year | 2018: 0.99 · 2019: 0.88 · 2020: 0.80 · 2021: 0.68 · 2023: 0.23 · 2024: 0.06 |
| **Blind test, 2025 fires** (39 cells) | ROC-AUC **0.25** · 0 % of fire cells in the alert zones |

- **Works** on the large fire years (2018–2021).
- **Breaks** on 2023–2025: the 2025 fires started in cells rated low risk.
- **Why:** the strongest features (elevation, recent burns, aspect) describe terrain and history, not **vegetation dryness** or **human access**.
- **Few positives:** only 345 fire cell-years, so the scores per year are unstable.
- No leakage: fire history strictly before each date; 2025 never used for training or model selection.

IMPACT

# Earlier, local, understandable fire information

- **SDG 13 Climate action** (13.1): resilience to climate hazards in a region where heatwaves and *chergui* winds drive extreme fire seasons.
- **SDG 15 Life on land** (15.1, 15.2): patrols and firebreak work targeted where the risk is highest.
- **SDG 11 Sustainable communities** (11.5): early warning for the ~10,600 people living inside the forest.
- **Morocco:** supports the national strategy *Forêts du Maroc 2020-2030*, ANEF and Civil Protection.
- **Arab region:** built only on open data (FIRMS, Sentinel-2, Landsat, DEM, WorldPop, Open-Meteo). It can be replicated in the Mediterranean forests of Algeria, Tunisia, Lebanon and Syria.
- **Inclusive:** Arabic-first, right-to-left interface, SMS for residents without smartphones.

NEXT STEPS FOR INCUBATION

# From proof of concept to an operational service

1. **Better predictors:** add land cover, OpenStreetMap distances and daily Sentinel-2 NDVI / NDMI / NBR from the research notebook to the reproducible pipeline.
2. **Hyperspectral:** EnMAP / Arab Satellite 813 maps of live fuel moisture and species.
3. **Better truth:** validate on dNBR burned areas, not only FIRMS points.
4. **Operations:** SMS gateway, a daily batch job, and an alert workflow with forest managers.
5. **Pilot:** one fire season in Bouhachem, Derdara and Jbel Sougna, then other Mediterranean forests.

Repository

**github.com/SanaeBelfrouh/wiqayagit**

Reproduce the headline result

```
pip install -r requirements.txt
jupyter lab notebooks/02_main_analysis.ipynb
```